

SEC P vs NP — notebook summary

SEC (autonomous) — attributed to Ludovico Kubler

2026-04-27 16:44 UTC

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Generated 2026-04-27 16:44 UTC

- Cycles recorded: **158**
- Time span: 91.1h (~1.73 cycles/h)
- Notebook: /home/ludo/Scrivania/SEC/research/pvsnp_notebook.jsonl

Verdict distribution

Verdict	Count
INCONCLUSIVE	139
FALSIFIED	14
SUPPORTED	4
BARRIER_HIT	1

Mathematical fields explored (field_A)

Field	Cycles
Ergodic Circuit Framework (communication complexity via dynamical systems)	4
{“framework_name”: “Ergodic Circuit Framework”, “math_branch”: “COMM_COMPLEXITY”}	4
Fourier analysis of boolean functions	3
Noncommutative geometry	2
Representation theory of symmetric groups	2
Ergodic Circuit Framework (COMM_COMPLEXITY)	2
Matroid Theory	2
Algebraic Topology	2
Tropical geometry	2
Random Matrix Theory	2
Combinatorial homotopy theory	1
Kazhdan-Lusztig theory of Hecke algebras	1
Combinatorial algebraic topology	1
Knot theory (quantum invariants)	1
Quadratic forms over finite fields (Grothendieck-Witt groups)	1
Clifford algebras over finite fields	1
Toric geometry (via Gröbner degenerations)	1

Field	Cycles
Arithmetic geometry (Tate-Shafarevich groups)	1
Geometric Langlands correspondence	1
Algebraic graph theory (chromatic polynomials)	1
Clifford algebras over real vector spaces	1
Algebraic cycles (Chow groups over finite fields)	1
Algebraic lattice theory (Möbius functions of flow lattices)	1
Directed algebraic topology (d-space homology)	1
Motivic integration (Denef-Loeser zeta functions)	1
Matroid theory (Clifford index of binary matroids)	1
Representation theory of finite groups (Frobenius-Schur indicator)	1
Categorification and knot Floer homology	1
Modular forms	1
Algebraic geometry (ideals in polynomial rings)	1

Last 15 cycles (chronological)

Time	Verdict	Title
2026-04-27 10:47 UTC	INCONCLUSIVE	Permanent of Sensitive-Boundary Bipartite Matrix Bounds Lifted In
2026-04-27 11:13 UTC	INCONCLUSIVE	Subdeterminant Dispersion Lower-Bounds Singular-Tail Matrix Rigid
2026-04-27 11:46 UTC	INCONCLUSIVE	Lyndon Factor Count of Truth Table Lower-Bounds DT Leaves Under I
2026-04-27 12:12 UTC	INCONCLUSIVE	Kashin-Split Energy Gap Lower-Bounds Sign-Matrix Rigidity at Rank
2026-04-27 12:41 UTC	INCONCLUSIVE	Roe-trace pairing for Inner-Product realizes coarse depth $\kappa \geq \log$
2026-04-27 12:47 UTC	INCONCLUSIVE	Cubical Betti Sum of Implicant Complex Lower-Bounds DNF-MCSP
2026-04-27 13:16 UTC	INCONCLUSIVE	Zeckendorf Length of NW Design Rows Bounds PRG Parity Bias
2026-04-27 13:49 UTC	INCONCLUSIVE	Sturmian Factor-Complexity Lower-Bounds DNF-MCSP Term Count
2026-04-27 14:18 UTC	INCONCLUSIVE	Haar-Wavelet Star-Discrepancy of Sign Matrix Lower-Bounds Rigidit
2026-04-27 14:43 UTC	INCONCLUSIVE	Roe-cohomological concentration: $HX^1(X_f)$ vanishes for random f ,

Time	Verdict	Title
2026-04-27 14:50 UTC	INCONCLUSIVE	Antichain-Width of Query Certificate Poset Bounds Lifted IND CC
2026-04-27 15:25 UTC	INCONCLUSIVE	Apolar Span of Clause-Product Polynomial Bounds DPLL Leaves
2026-04-27 15:55 UTC	INCONCLUSIVE	Cheeger Constant of Quotient Multigraph Bounds Tseitin Resolution
2026-04-27 16:27 UTC	INCONCLUSIVE	RSK Second-Row Length of f-Sorted Permutation Bounds Lifted IND C
2026-04-27 16:44 UTC	INCONCLUSIVE	Coarse-dimensional KRW direct sum: asdim is sub-additive under bl

How to read the reports

- reports/supported_findings.md — SUPPORTED conjectures with full test code. Paper-quality.
- reports/falsified_findings.md — FALSIFIED with counterexamples.
- reports/daily_YYYY-MM-DD.md — chronological log by day.
- pvsnp_notebook.jsonl — raw JSONL, one entry per cycle (source of truth).

PDF export

Install pandoc + a LaTeX engine once (`sudo apt install pandoc texlive-xetex`), then:

```
cd research/reports
for f in supported_findings.md falsified_findings.md notebook_summary.md; do
    pandoc "$f" -o "${f%.md}.pdf" --pdf-engine=xelatex -V geometry:margin=2cm
done
```